

Sprint 4 Plan

SlugRoute

{Week 10: 6/03/26 11:59pm}

Revision 1 {5/16/2026}

Burn Down Chart

 **Sprint 4 Burn Down Chart**

Scrum Board

 **Sprint 4 Scrum Board**

Screenshots

 **SlugRoute Screenshots**

Goal:

Finalize the student "Daily Companion" experience by implementing automated sequential routing between back-to-back classes, securing the API infrastructure, and enabling calendar synchronization for offline schedule access.

User Stories

1. As a student, I want to see my entire day's walking route so I can navigate my back-to-back classes efficiently. 13

1. Chronological Schedule View: Refactor the "Saved for Later" sidebar into a daily itinerary that automatically groups and sorts saved courses by time. (6)
2. Sequential Waypoint Logic: Research and implement the logic to handle multiple waypoints (Class A -> Class B) within a single route request. (4)
3. Multi-path Rendering: Update the frontend map component to render continuous paths between consecutive waypoints. (4)
4. Walk Estimates: Use the Google Routes API to calculate the time and distance for each individual route (for multiple or single meetings). (4)

2. As a student, I want to trust that the map data is always accurate and my usage is secure so that I never miss a class due to an outdated schedule. 5

1. Automated Data Synchronization: Implement a GitHub Action or local Cron job to trigger scraper.py every 24 hours to reflect PISA updates. (6)
2. Backend API Proxy & Security: Move Google Maps API Keys to environment variables and implement a Go proxy endpoint to protect the application from unauthorized access. (6)

3. As a student, I want to export my mapped schedule to my phone's calendar so that I can view my room numbers and locations while offline. 8

1. RFC 5545 Data Schema: Define the Go structs required to generate valid iCalendar (.ics) files, including UIDs and timezone offsets. (4)
2. iCalendar Stream Generator: Develop the Go service that marshals SQLite course records into the final .ics text stream. (4)
3. UCSC Term Recurrence Logic: Implement logic to handle UCSC's recurring weekly patterns (MWF/TTh) and exclude holidays/finals week based on the academic calendar. (6)
4. Frontend Export & FileStream Integration: Add the "Sync to Calendar" button and JavaScript logic to trigger the native browser file-save dialog for the .ics file. (6)

4. As a mobile user, I want a responsive, polished UI that persists my data reliably across sessions. 5

1. Mobile Navigation Optimization: Refactor the sidebar and search preview to ensure they are accessible and legible for one-handed use on small screens. (4)
2. Touch Target & CSS Audit: Conduct a comprehensive audit to ensure all interactive elements meet mobile spacing standards and dark mode consistency. (4)
3. State Hardening: Harden LocalStorage logic to ensure saved courses and visibility toggles persist across browser updates or cache clears. (5)
4. Final Branding: Finalize custom iconography and Sammy the Slug map markers. (3)

Total Hours: 66

Team Roles

jdcody@ucsc.edu :	Developer
jlores@ucsc.edu :	Developer, Product Owner
fveloz@ucsc.edu :	Developer
avaldesq@ucsc.edu :	Developer

Initial task assignment

jdcody@ucsc.edu :	User Story 2, Task 1
jlores@ucsc.edu :	User Story 1, Task 3
fveloz@ucsc.edu :	User Story 4, Task 2
avaldesq@ucsc.edu :	User Story 3, Task 3

Scrum times

TA meeting: Monday (10am-11am)
Team meeting 1: Tuesday (5pm-7pm)
Team meeting 2: Thursday (5pm-7pm)
Team meeting 3: Sunday (12pm-2pm)